

Sales Performance Analysis and Business Insights Dashboard Using Microsoft Excel

Excel Mini Project Report

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Course: Data Analytics

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Introduction:

Sales data plays an important role in helping businesses understand their performance and make informed decisions. This project, **"Sales Performance Analysis and Business Insights Dashboard Using Microsoft Excel,"** was carried out to apply Excel-based analytical techniques for processing and visualizing retail sales data. The project covers the complete workflow from data preparation to dashboard creation, providing practical experience in data analysis, visualization, and business reporting using Microsoft Excel.

Problem Statement:

ABC Electronics maintains a large volume of sales records from different regions and product categories. Since the data is stored in raw format, it is difficult to identify important trends, compare regional performance, or evaluate product category sales efficiently. To overcome this challenge, the dataset was cleaned, analyzed using Excel formulas and Pivot Tables, and transformed into an interactive dashboard that presents key business metrics and visual insights for easier analysis and decision-making.

Objectives:

The main objectives of this project were:

- To clean and standardize the sales dataset using Excel functions.
- To practice searching and filtering records based on specific conditions.
- To perform calculations using Excel formulas for business analysis.
- To summarize large datasets using Pivot Tables.
- To create different charts for effective data visualization.
- To develop an interactive dashboard with KPIs, charts, and slicers for business reporting.

Dataset Description:

The project uses a sales dataset containing approximately **4,000 records** collected from ABC Electronics.

Dataset Columns:

Column Names	Description
Order ID	Unique identifier for each order
Order Date	Date on which the order was placed
Region	Sales region
Product Category	Category of the product sold
Sales Amount	Total sales value
Quantity Sold	Number of units sold
Discount (%)	Discount offered on the sale
Profit	Profit earned from the transaction

Project Workflow:

The project was completed in several stages, starting from data preprocessing and ending with dashboard development.

➤ Data Cleaning:

The raw dataset contained inconsistent text formatting and unnecessary spaces, which could affect the accuracy of the analysis. To standardize the data, Excel text functions were used.

- **TRIM ()** function was used to remove unnecessary spaces from next station.
- The **UPPER ()** and **LOWER ()** functions were used to make the **Region** and **Product Category** columns became consistent throughout the dataset.
- I used **SUBSTITUTE** and **FIND&REPLACE** functions to correct the inconsistent data in region and category.

➤ Searching and Filtering:

The dataset was filtered using Excel's Filter feature to display orders from the **South** region for the **Electronics** category within the last year, enabling focused analysis.

➤ Data Validation:

The dataset was validated by checking for blank values, ensuring consistent formatting, and identifying duplicate records to maintain data accuracy and consistency.

➤ Formula Calculations:

Excel formulas such as **SUM ()**, **AVERAGEIF ()**, **FILTER ()** were used to calculate Total Sales, Total Profit, Average Discount, Average Profit, and Profit Margin for dashboard reporting.

SUM (): Used to calculate the total sales for each region.

AVERAGEIFS (): Used to calculate the average discount and average profit for the furniture category.

FILTER (): Used to specific the records.

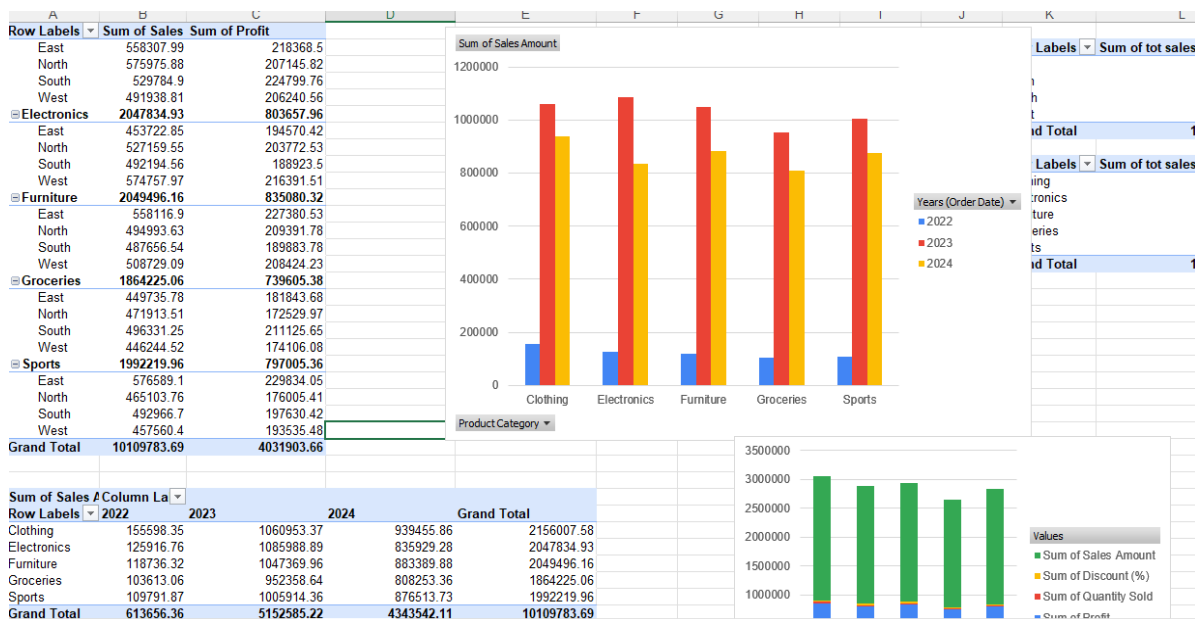
➤ Merging Regional Average Sales:

The **AVERAGEIFS ()** function was used to calculate the average sales for each region and merge the values back into the dataset for regional comparison.

➤ Pivot Table Analysis:

Pivot Tables were created to summarize sales and profit by region and product category, with slicers added for interactive filtering.

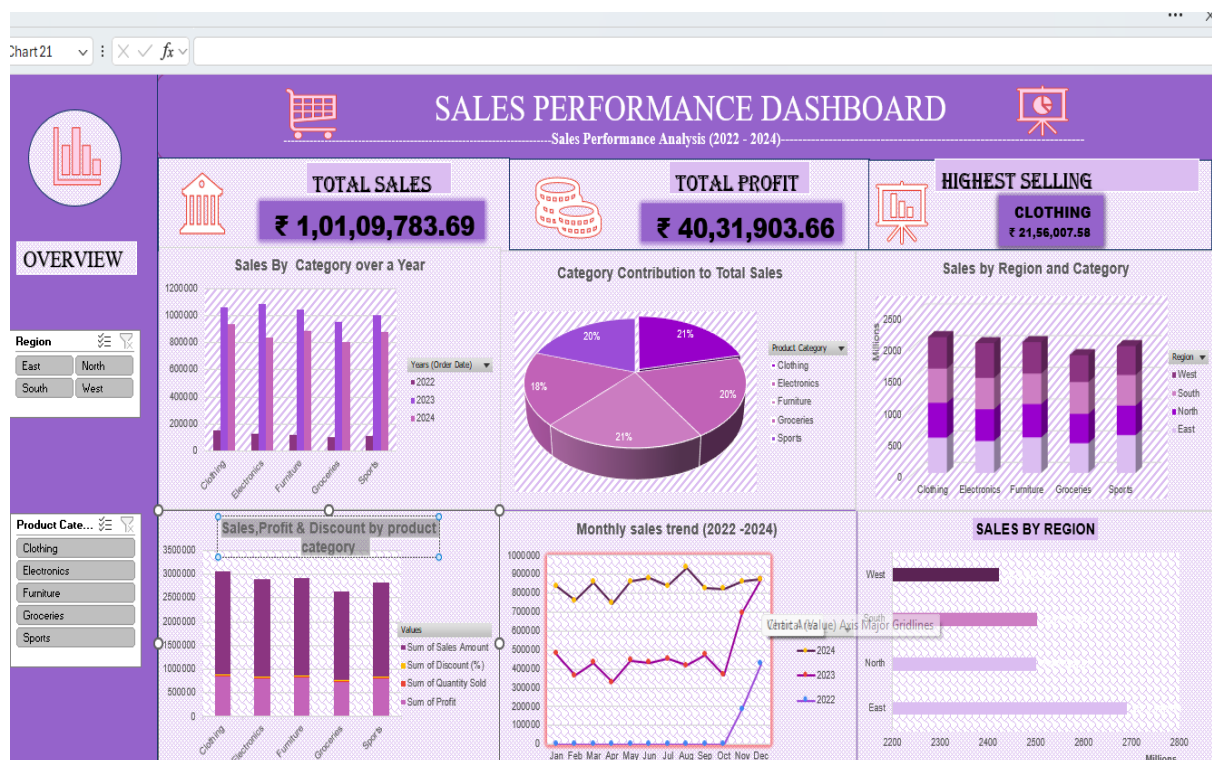
- Sales by category over a region.
- Category Contribution to Total Sales
- Sales by Region and Category
- Sales by Region
- Sales, Profit & Discount by product category.



- Slicers were connected to the Pivot Tables, allowing users to filter the data dynamically and view different summaries without modifying the original dataset. This made the analysis more interactive and user-friendly.

➤ Dashboard:

The dashboard includes KPIs such as **Total Sales**, **Total Profit**, and **Highest Selling Product Category**. It also contains charts that display sales trends, regional performance, and product category contributions. Interactive slicers for **Region** and **Product Category** were added to enable dynamic filtering. This dashboard provides a clear, user-friendly interface for analyzing sales performance and business insights.



➤ Conditional Formatting:

Conditional Formatting was applied to highlight important sales records automatically. Transactions with a **Profit Margin greater than 50%** and **Sales Amount exceeding ₹4,000** were highlighted using formatting rules. This feature helps users quickly identify high-performing transactions and makes the dataset easier to analyze visually.

Business Insights

1. Strong Revenue Growth

Total sales exceed **₹1 crore**, indicating healthy business performance.

2. Healthy Profitability

Profit exceeds **₹40 lakh**, reflecting effective pricing and cost management.

3. Continuous Yearly Growth

2024 outperforms both 2022 and 2023 across almost all categories, showing positive business expansion.

4. Balanced Product Portfolio

Revenue is well distributed across product categories, reducing dependence on any single product line.

5. Seasonal Sales Pattern

Sales peak toward the end of the year, suggesting festive or holiday demand.

6. Regional Opportunities

East is the strongest-performing region, while West has room for improvement through targeted sales initiatives.

Business Recommendations

1. Increase inventory for Clothing, Furniture, and Sports during peak sales periods.
2. Strengthen marketing efforts for the Groceries category to improve its contribution.
3. Launch region-specific promotions, particularly to boost sales in the West.
4. Optimize discount strategies to protect profit margins while supporting slow-moving products.
5. Prepare additional stock and logistics before year-end to capitalize on seasonal demand.
6. Continue analyzing customer behaviour by region and category to enable personalized marketing and improve customer retention.

Conclusion:

The **Sales Performance Dashboard** shows consistent business growth, with **total sales of ₹1.01 crore** and **total profit of ₹40.31 lakh**. Sales increased steadily from **2022 to 2024**, with **2024 achieving the highest performance**. The product categories contribute evenly to revenue, while the **East region** performs best and the **West region** has scope for improvement. Overall, the dashboard supports effective business decision-making and future growth.